

Postoperative Pain After the use of Sodium Hypochlorite gel and Solution Forms: A Randomized Clinical Study

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HIGHLIGHTS

- This the first study comparing NaOCl gel and solution forms in terms of postoperative pain
- It has been revealed that NaOCl gel during root canal preparation results in less postoperative on day 1.
- This is the first study revealing the beneficial effect of NaOCl gel in lowering postoperative pain

INTRODUCTION

Irrigation of the root canals is an essential part of endodontic treatment because the complete elimination of microorganisms from the root canal system is impossible only with instrumentation (1). Sodium hypochlorite (NaOCl) is the most common irrigant used in endodontics because of its physicochemical and antibacterial properties (2), and its unique ability to dissolve necrotic tissue

remnants (3). However, accidental injection of the solution into the periapical tissues can result in some serious complications such as; hemolysis, ulceration, allergic reaction and tissue necrosis (4). To avoid complications associated with extrusion of the solution, gel form of the NaOCl solution can be proposed. Previously, it has been reported that the gel and solution both did not interfere in the EDTA solution's action (5). Additionally, Garcia et al. (6) have evaluated the effect of several forms of NaOCl on the microhardness of root canal dentin and reported that the NaOCl gel and solution forms have a similar effect on dentinal microhardness. Moreover, the antibacterial effectiveness of NaOCl gel and solution forms were compared and it has been reported that the antibacterial efficacy of the NaOCl gel and solution forms were similar (7). However, the tissue dissolution capacity of the NaOCl solution was higher than the gel form (7).

Another complication is postoperative pain that occurs when the solution extruded into the periapical tissues (8). Several strategies have been recommended to reduce postoperative pain include medication (9), intracanal cryotherapy (10), low-level laser therapy (11), occlusal reduction (12), and glide-path application (13). However, to date, the effect of using NaOCl gel for root canal disinfection during root canal preparation on postoperative pain has not been studied. Thus, the present study aimed to compare the effect of NaOCl gel and solution forms on postoperative pain

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level. The null hypothesis was that there would be no difference between the NaOCl gel and solution groups in terms of postoperative pain.

MATERIALS AND METHODS

The Ethical Committee of the Faculty of Dentistry, University of A approved the study protocol with a decision number of 2017-91. All the participants included in the study signed an informed consent form before undergoing the treatment. The sample size calculation, which was made by a program (GPower; Franz Faul, University of Kiel, Germany) according to the data obtained from a previous study (11), indicated that 18 patients were sufficient for per group with an error of $\alpha=0.05$, effect size of 0.84 and power of 0.8. However, considering the possible loss of the participants and to increase the statistical power, 30 patients for each group were included. Randomisation of the participants was performed by using a web program (www.randomizer.org) and all the patients were blinded to the groups.

Healthy patients (American Society of Anesthesiology [ASA] I) having molar teeth diagnosed with a periapical diagnosis of symptomatic apical periodontitis and a pulpal diagnosis of symptomatic irreversible pulpitis without radiolucency and with a preoperative pain of more than 50 on 10 cm visual analog scale (VAS) were included to the study. Electric pulp tester (EPT) (Digitest, Parkell, Edgewood, NY, USA) and cold test (Roeko Endo Frost, Coltene, Langenau, Germany) were used to test the sensibility of the teeth. Also, a percussion test was performed using finger and end of an instrument. Exclusion criteria were as follows: teeth having previous root canal treatment, root fracture, swelling, pathological mobility, a pocket depth of more than 3 mm and patients who are older than 65 years old or younger than 18, and taking antibiotics or analgesic within one month prior to the study. Teeth having radiographic lesions were excluded. Only teeth having healthy or widened periodontal ligament were included.

All the teeth had been anaesthetised using 1.8 ml 4% articaine with 1:100.000 epinephrine (Ultracaine DS Forte; Aventis, Istanbul, Turkey) before the isolation of the teeth with rubber dam. For maxillary molars, local infiltration anaesthesia was performed and for mandibular molars, mandibular anaesthesia was performed. Access cavities were prepared and the coronal thirds of the root canals were instrumented using Reciproc R25 files (VDW GmbH, Munich, Germany). Following root canal irrigation with 2 mL of 5.25% NaOCl (Chloraxid, Cer-kamed, Poland), the working lengths were determined using an apex locator (Raypex 6, VDW). Then, all the root canals were prepared using Reciproc instruments according to the manufacturer's instructions at full working length.

NaOCl solution group: During preparation, the root canals were irrigated with 2 mL of 5.25% NaOCl (Chloraxid, Cer-kamed) between each three pecking motion.

NaOCl gel group: During preparation, 5.25% NaOCl gel (Chloraxid, Cer-kamed) was used according to the manufacturer's instructions. The NaOCl gel was applied directly into the canal by using the applicator and root canal preparation was performed. The root canals were then flushed with 2 mL of saline.

The amount of NaOCl gel applied into the root canal between each pecking motion was 2 mL. This procedure was repeated between each three pecking motion until the root canal preparation was completed.

The root canals were then irrigated with 6 mL of 17% EDTA for the final irrigation once the root canal preparation was completed. The root canals were dried with paper points and obturated with gutta-percha cones and sealer (Sealapex, Kerr Corporation, Orange, CA) using cold lateral condensation technique. Then the permanent restorations were performed with resin composite (3M ESPE, St Paul, MN). The participants were given instructions to record postoperative pain levels on 24, 48, and 72 hours and 1 week after treatment using VAS and analgesic taken on the questionnaire. In case of the patient was referred to an unscheduled appointment, it was also recorded.

CONCLUSION

Within the limitations of the present study, it can be concluded that using NaOCl gel during root canal preparation results in less postoperative on day 1. This is the first study revealing the beneficial effect of NaOCl gel in lowering postoperative pain. Thus, further studies are needed.

Disclosures

Conflict of interest: There are no any conflict of interest.

Ethics Committee Approval: The Ethical Committee of the Faculty of Dentistry, University of A approved the study protocol with a decision number of 2017-91.

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